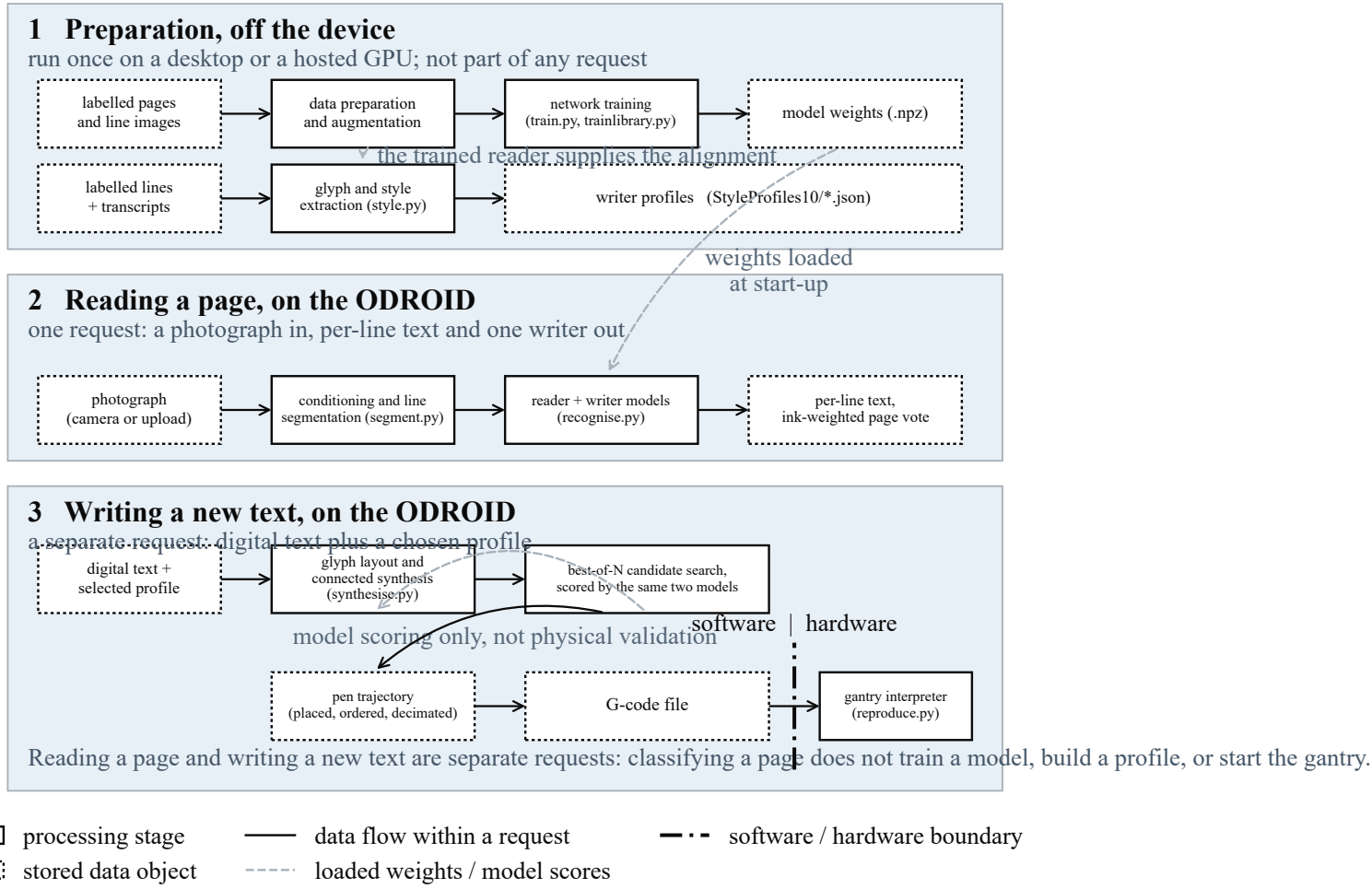

1 Preparation, off the device
run once on a desktop or a hosted GPU; not part of any request
labelled pages and line images
data preparation and augmentation
network training (train.py, trainlibrary.py)
model weights (.npz)
the trained reader supplies the alignment
labelled lines + transcripts
glyph and style extraction (style.py)
writer profiles (StyleProfiles10/*.json)
weights loaded at start-up
2 Reading a page, on the ODROID
one request: a photograph in, per-line text and one writer out
photograph (camera or upload)
conditioning and line segmentation (segment.py)
reader + writer models (recognise.py)
per-line text, ink-weighted page vote
3 Writing a new text, on the ODROID
a separate request: digital text plus a chosen profile
digital text + selected profile
glyph layout and connected synthesis (synthesise.py)
best-of-N candidate search, scored by the same two models
model scoring only, not physical validation
software | hardware
pen trajectory (placed, ordered, decimated)
G-code file
gantry interpreter (reproduce.py)
Reading a page and writing a new text are separate requests: classifying a page does not train a model, build a profile, or start the gantry.
processing stage
stored data object
data flow within a request
loaded weights / model scores
software / hardware boundary